

TUGAS 2

Date _____

Tugas 2

1. $y = 3x^2 - 2x + 5$

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{3(x+h)^2 - 2(x+h) + 5 - (3x^2 - 2x + 5)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{3x^2 + 6xh + 3h^2 - 2x - 2h + 5 - 3x^2 + 2x - 5}{h}$$

$$= \lim_{h \rightarrow 0} \frac{6xh + 3h^2 - 2h}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(6x + 3h - 2)}{h}$$

$$= \lim_{h \rightarrow 0} 6x + 3 \cdot 0 - 2$$

$$\frac{dy}{dx} = 6x - 2$$

2. $f(x) = x^2 - 1$

a) Persamaan G. Singgung

$$f(x) = x^2 - 1$$

$$f'(x) = 2x$$

Dika Titik Singgung $x = a$

$$f(a) = a^2 - 1, \text{ Jadi titiknyanya } (a, a^2 - 1)$$

$$y - y_1 = m(x - x_1)$$

$$y - (a^2 - 1) = 2a(x - a)$$

$$y = 2a(x - a) + (a^2 - 1)$$

b) Persamaan G. Normal

$$M = a$$

$$M_{\text{normal}} = -\frac{1}{2a}$$

$$y - y_1 = m(x - x_1)$$

$$y - (a^2 - 1) = -\frac{1}{2}(x - a)$$

$$y = -\frac{1}{2}(x - a) + (a^2 - 1)$$

c)

(=) G. Singgung dan Normal yang potongannya di garis $x = -1$

$$x = -1$$

$$F(x) = x^2 - 1 \Rightarrow y = (-1)^2 - 1 \Rightarrow (-1, 0)$$

$$f'(x) = 2x \Rightarrow y = 0$$

$$M = f'(-1)$$

$$= 2 \cdot (-1)$$

$$M = -2$$

$$y = mx + c$$

$$0 = -2x + c$$

$$-2x + c = 0$$

$$c = 0 + 2x$$

$$c = 0 + 2 \cdot (-1)$$

$$c = -2$$

$$y - y_1 = m(x - x_1)$$

$$y - 0 = -2(x + 1) \quad \{G. \text{ Singgung} \}$$

$$y = -2x - 2$$

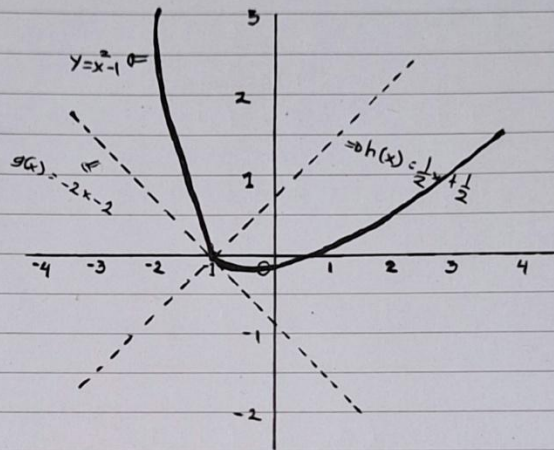
$$M_{\text{normal}} = -\frac{1}{m}$$

$$\Rightarrow y - y_1 = m(x - x_1)$$

$$y - 0 = \frac{1}{2}(x + 1)$$

$$y = \frac{1}{2}x + \frac{1}{2} \quad \{G. \text{ Normal} \}$$

⇒ Gambar Kurva :



3. $y = x \cos 2x$

a) $y'''(x) \dots ?$

$$u = x$$

$$\Rightarrow u' = 1$$

$$v = \cos 2x$$

$$\Rightarrow v' = -2 \sin(2x)$$

$$y' = u' \cdot v + u \cdot v'$$

$$= 1 \cdot \cos(2x) + x \cdot (-2 \sin(2x))$$

$$= \cos 2x - 2x \sin 2x$$

$$u = 2x$$

$$\Rightarrow u' = 2$$

$$v = \sin 2x$$

$$\Rightarrow v' = 2 \cos(2x)$$

$$y'' = -2 \sin 2x - (2 \sin 2x + 2x \cdot 2 \cos 2x)$$

$$= -2 \sin 2x - 2 \sin 2x - 4x \cos 2x$$

$$= -4 \sin 2x - 4x \cos 2x$$

$$u = -4x \quad \Rightarrow \quad u' = -4$$

$$v = \cos 2x \quad \Rightarrow \quad v' = -2 \sin 2x$$

$$y''' = -4(2 \cos 2x) - (y \cdot \cos 2x + yx(-2 \sin 2x))$$

$$= -8 \cos 2x - 4 \cos 2x + 8x \sin 2x$$

$$= -12 \cos 2x + 8x \sin 2x$$

$$b) y'''(\pi) = -12 \cos 2(\pi) + 8(\pi) \sin 2(\pi)$$

$$= -12 \cdot (1) + 8 \cdot (0)$$

$$= -12$$

4.

$$a) y = \ln(\sin(x^3-1))$$

$$\ln = \frac{1}{\sin(x^3-1)}$$

Gunakan Aturan Rantai:

$$\sin(x^3-1) = \cos(x^3-1) \cdot 3x^2$$

$$\frac{1}{\sin(x^3-1)}$$

$$\frac{dy}{dx} = \frac{1}{\sin(x^3-1)} \cdot \cos(x^3-1) \cdot 3x^2$$

$$\frac{dy}{dx} = \frac{3x^2 \cos(x^3-1)}{\sin(x^3-1)} \quad \text{atau} \quad 3x^2 \cot(x^3-1)$$

$$b) f(x,y) = 2x^2y - \sin xy^2 + e^{-xy} = 10$$

$$\Rightarrow 4x \frac{dx}{dx} \cdot (y) + 2x^2 \cdot 1 \frac{dy}{dx} - \cos(xy^2) \cdot ($$

$$1 \frac{dx}{dx} \cdot (y^2) + x(2y \frac{dy}{dx}) + e^{-xy}(-1(y+x \cdot 1 \frac{dy}{dx})) = 0$$

$$\Rightarrow 4xy + 2x^2 \frac{dy}{dx} - \cos(xy^2) \cdot y^2 - 2xy \cdot \cos(xy^2) \frac{dy}{dx}$$

$$-e^{-xy} \cdot y - x \cdot e^{-xy} \frac{dy}{dx} = 0$$

↓

$$\Rightarrow (2x^2 - 2xy \cos(xy^2) - x \cdot e^{-xy}) \frac{dy}{dx} = -4xy + y^2 \cos(xy^2) + y \cdot e^{-xy}$$

$$\frac{dy}{dx} = \frac{-4xy + y^2 \cos(xy^2) + y \cdot e^{-xy}}{2x^2 - 2xy \cos(xy^2) - x \cdot e^{-xy}}$$

5. $f(x) = \sin 2x$

a) Deret Taylor $x = \frac{\pi}{2}$, Selama 5 suku ;

$$\Rightarrow f(x) = \sin 2x \quad \Rightarrow \quad f\left(\frac{\pi}{2}\right) = \sin 2\left(\frac{\pi}{2}\right)$$

$$= \sin \pi$$

$$= 0$$

$$\Rightarrow f'(x) = 2 \cos 2x \quad \Rightarrow \quad f'\left(\frac{\pi}{2}\right) = 2 \cos 2\left(\frac{\pi}{2}\right)$$

$$= 2 \cos \pi$$

$$= -2$$

$$\Rightarrow f''(x) = 2(-2 \sin 2x) \quad \Rightarrow \quad f''\left(\frac{\pi}{2}\right) = -4 \sin\left(2 \cdot \frac{\pi}{2}\right)$$

$$= -4 \sin(2x)$$

$$= -4 \cdot 0$$

$$= 0$$

$$\Rightarrow f'''(x) = -8 \cos 2x \quad \Rightarrow \quad f'''\left(\frac{\pi}{2}\right) = -8 \cos 2\left(\frac{\pi}{2}\right)$$

$$= -8 \cdot (-1)$$

$$= 8$$

$$\begin{aligned} \text{e)} \quad f'''(x) &= -8 (-2 \sin 2x) \Rightarrow f'''(\frac{\pi}{2}) = 16 \sin 2(\frac{\pi}{2}) \\ &= 16 \sin(2x) \end{aligned}$$

$$\begin{aligned} &= 16 \sin \pi \\ &= 16 \cdot 0 \\ &= 0 \end{aligned}$$

Jaki Deret Taylor dla $x = \frac{\pi}{2}$

$$= 0 + \frac{(-2)}{1!} (x - \frac{\pi}{2}) + \frac{0}{2!} (x - \frac{\pi}{2})^2 + \frac{8}{3!} (x - \frac{\pi}{2})^3 + \frac{0}{4!} (x - \frac{\pi}{2})^4$$

$$= \frac{-2}{1} (x - \frac{\pi}{2}) + (x - \frac{\pi}{2})^2 + \frac{4}{3} (x - \frac{\pi}{2})^3 + (x - \frac{\pi}{2})^4$$

$$= -2 (x - \frac{\pi}{2}) + (x - \frac{\pi}{2})^2 + \frac{4}{3} (x - \frac{\pi}{2})^3 + (x - \frac{\pi}{2})^4$$

b) Deret Maclaurin

$$\begin{aligned} \text{e)} \quad f(x) &= \sin 2x & \Rightarrow f(0) &= \sin 2(0) \\ & & &= \sin 0 \\ & & &= 0 \end{aligned}$$

$$\begin{aligned} \text{e)} \quad f'(x) &= 2 \cos 2x & \Rightarrow f'(0) &= 2 \cos 2 \cdot 0 \\ & & &= 2 \cos 0 \\ & & &= 2 \cdot 1 \\ & & &= 2 \end{aligned}$$

$$\begin{aligned} \text{e)} \quad f''(x) &= -4 \sin 2x & \Rightarrow f''(0) &= -4 \sin 2 \cdot 0 \\ & & &= -4 \sin 0 \\ & & &= -4 \cdot 0 \\ & & &= 0 \end{aligned}$$

$$\begin{aligned} \text{e)} \quad f'''(x) &= -8 \cos 2x & \Rightarrow f'''(0) &= -8 \cos 2 \cdot 0 \\ & & &= -8 \cos 0 \\ & & &= -8 \end{aligned}$$

$$\begin{aligned} \text{e)} \quad f^{(4)}(x) &= 16 \sin 2x & \Rightarrow f^{(4)}(0) &= 16 \sin 2 \cdot 0 \\ & & &= 16 \cdot 0 \\ & & &= 0 \end{aligned}$$

No. _____

Date _____

$$F(x) = 0 + \frac{2x}{1!} + \frac{0}{2!} x^2 + \frac{(-8)}{3!} x^3 + 0$$

$$= 2x + \frac{(-8)}{6} x^3$$

$$= 2x - \frac{4}{3} x^3$$